

Auditing Model Substitution in LLM APIs

Why Software Tests Fail and Hardware Attestation Wins

arXiv:2504.04715 | ML Researchers • AI Security Practitioners • Cloud Infrastructure Engineers

Evidence-based analysis of the trust gap in commercial LLM APIs

The Model Substitution Problem

Economic Incentives Drive Covert Swaps in Commercial LLM APIs

Commercial LLM API providers charge premium rates for specific models (e.g., Llama-3-70B, Qwen2-72B), but the black-box nature of APIs means users have no verifiable guarantee the advertised model is actually running.

Providers face strong economic pressure to cut GPU costs by substituting cheaper alternatives — quantized weights, smaller architectures, or entirely different families — while maintaining pricing tiers.

Formal Hypothesis Testing Framework

H_0 : API serves $P_{\text{spec}}(y|x)$ vs. H_1 : API serves $P_{\text{actual}}(y|x) \neq P_{\text{spec}}(y|x)$

Four Substitution Vectors

① Smaller Model

Serving 8B instead of 70B parameter model; largest cost reduction

② Quantized Variant

FP8/INT8 quantized weights replacing FP16; marginal quality drop

③ Fine-tuned Version

Updated checkpoint or fine-tuned variant of the named model

④ Different Family

Entirely different model architecture served under same API name



Core challenge: all four vectors produce outputs that are statistically close to the specified model, especially at high temperature or with quantization — making discrimination query-intensive and unreliable.

FP8 Quantization Preserves Most Accuracy — Making Substitution Hard to Detect

Original vs. FP8-quantized benchmark scores across five model families

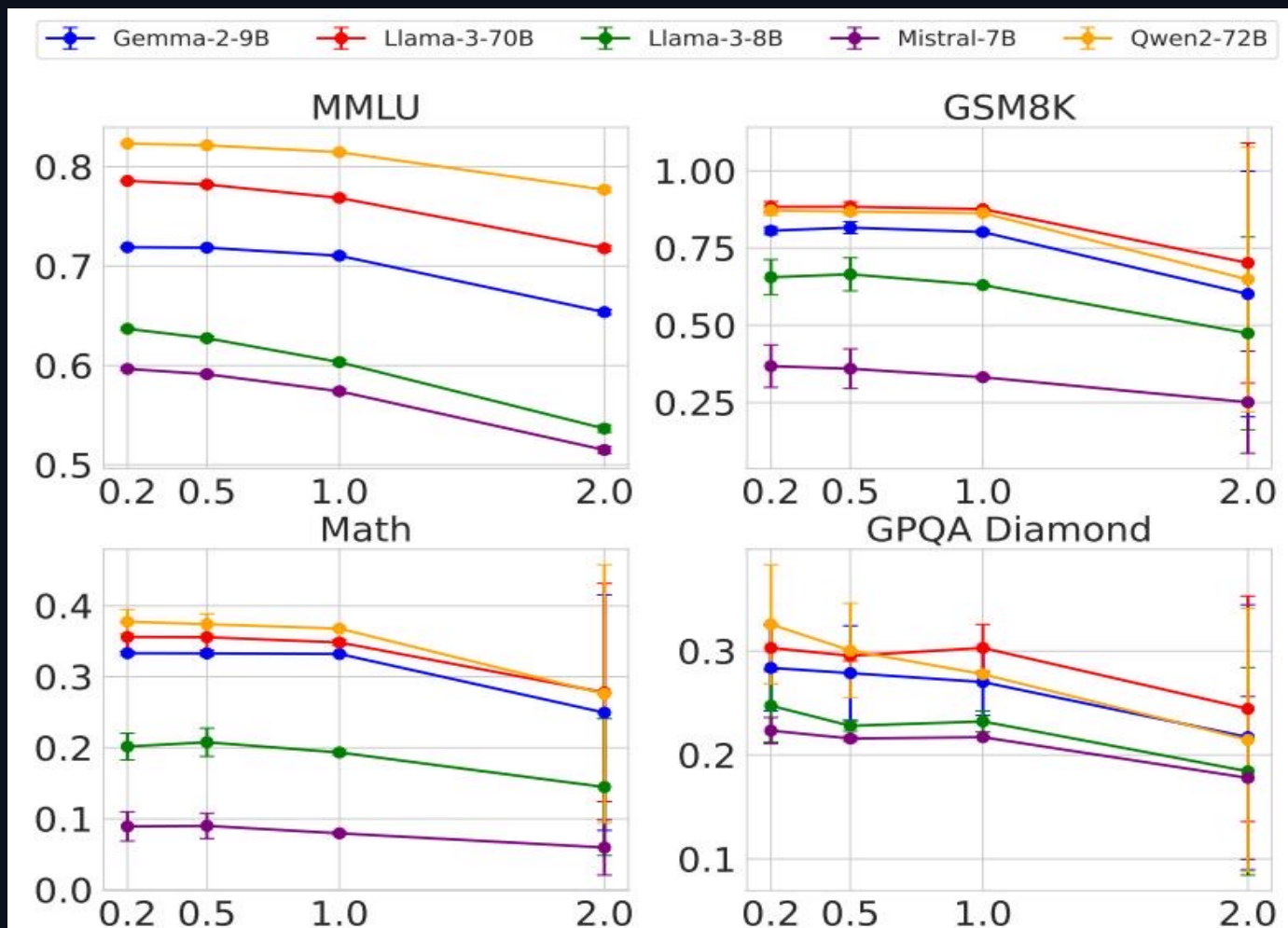
Model	Precision	MMLU	GSM8K	MATH	GPQA Diamond
Llama-3-8B-Instruct	FP16	62.69 ± 0.18	61.14 ± 3.47	20.65 ± 3.43	22.62 ± 0.22
Llama-3-8B-Instruct	FP8	62.43 ± 0.26	60.90 ± 4.10	14.91 ± 2.49	20.14 ± 0.24
Llama-3-70B-Instruct	FP16	78.05 ± 0.08	88.06 ± 1.44	35.69 ± 1.33	29.60 ± 0.51
Llama-3-70B-Instruct	FP8	77.88 ± 0.13	87.35 ± 1.24	35.75 ± 1.16	33.12 ± 0.30
Gemma-2-9b-it	FP16	71.86 ± 0.08	81.80 ± 1.35	33.41 ± 0.28	28.64 ± 2.97
Gemma-2-9b-it	FP8	71.92 ± 0.11	79.41 ± 1.14	32.53 ± 0.34	27.81 ± 3.22
Qwen2-72B-Instruct	FP16	82.18 ± 0.08	86.72 ± 1.00	37.39 ± 1.41	29.93 ± 2.71
Qwen2-72B-Instruct	FP8	81.98 ± 0.08	86.82 ± 0.97	37.67 ± 1.39	31.08 ± 1.96
Mistral-7B-Instruct-v0.3	FP16	59.15 ± 0.10	35.90 ± 4.54	8.94 ± 1.24	21.60 ± 0.17
Mistral-7B-Instruct-v0.3	FP8	58.77 ± 0.13	32.20 ± 4.02	7.68 ± 1.23	22.72 ± 0.19



FP8 scores fall within error bars of the original across most benchmarks — quantization enables significant cost reduction with near-indistinguishable quality, creating a strong economic incentive for covert substitution.

Higher Temperature Amplifies Performance Gaps but Increases Noise

Benchmark accuracy vs. temperature (0.2, 0.5, 1.0, 2.0) for all five model families



Key Observations

At $T = 0.2$

All models perform near their capability ceiling; original and quantized variants are nearly indistinguishable.

At $T = 0.5$

Performance begins to diverge, especially on GSM8K and MATH; some FP8 drops become visible.

At $T = 1.0$

Gaps widen across all tasks, but variance increases significantly — confidence intervals overlap substantially.

At $T = 2.0$

Performance degrades across the board; the gap between original and substitute narrows relative to noise, complicating detection.

▶ Detection via statistical tests at high temperature requires impractically large query budgets — noise dominates signal.

Log Probability Fingerprinting Fails Due to Inference Nondeterminism

Token-level log probabilities from the same Llama-3-8B model vary across software frameworks and GPU hardware

Model	BERT Acc	T5 Acc	GPT2 Acc	LLM2Vec Acc
Llama3-70B-Instruct-FP8	50.55	50.10	50.45	49.90
Llama3-70B-Instruct-INT8	51.60	49.90	51.30	50.25
Gemma2-9b-it-FP8	49.95	50.30	51.20	49.80
Gemma2-9b-it-INT8	49.00	49.70	51.65	49.55
Mistral-7b-v3-Instruct-FP8	50.55	49.75	48.70	48.75
Mistral-7b-v3-Instruct-INT8	49.70	50.75	50.50	51.15
Qwen2-72B-Instruct-FP8	50.05	50.25	48.75	49.80
Qwen2-72B-Instruct-INT8	50.75	50.55	49.45	50.20

Framework & Hardware Variation

transformers 4.40 → 4.50

Same model, same hardware; minor version bump produces measurably different token log-probs.

vLLM 0.6.2 → 0.8.2

Different batching and kernel paths introduce significant log-prob divergence.

A100 vs. H100

Hardware differences in floating-point rounding and memory layout further shift distributions.

▶ Production inference nondeterminism makes it impossible to distinguish a substituted model from infrastructure variation — log-probability fingerprinting is unreliable in practice.

Randomized Routing at Low Rates Makes Statistical Detection Query-Prohibitive

A provider can mix substitute responses with probability p , converging distributions as $p \rightarrow 0$

Attack Model

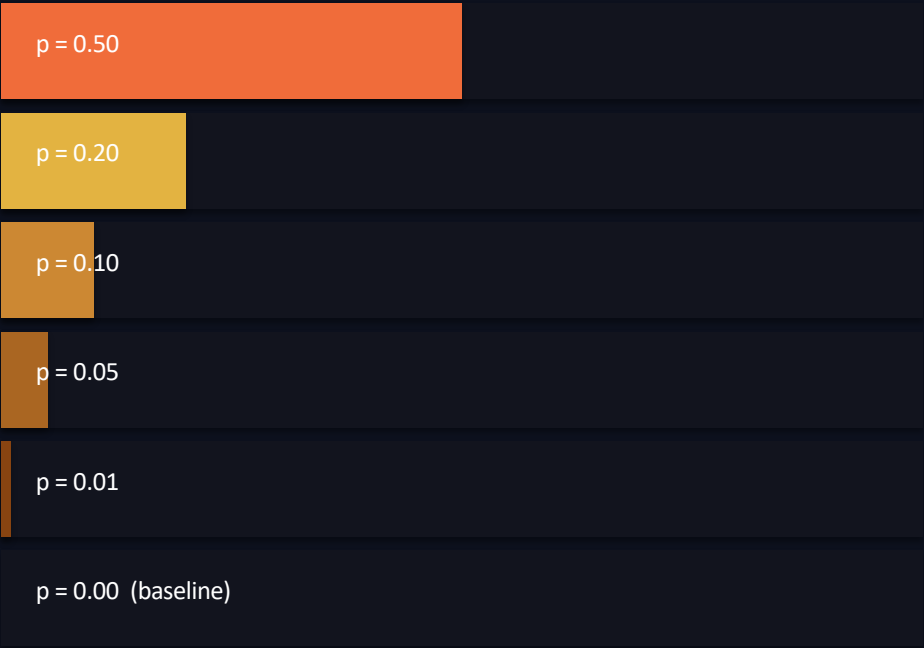
A adversarial provider routes a fraction p of queries to the cheaper substitute model and the remaining $(1-p)$ to the specified model. The mixed output distribution is:

$$P_{\text{mixed}}(y|x) = (1-p) \cdot P_{\text{spec}}(y|x) + p \cdot P_{\text{sub}}(y|x)$$

As $p \rightarrow 0$, P_{mixed} converges to P_{spec} — the auditor observes a distribution statistically indistinguishable from the specified model, even with thousands of queries.

Divergence from Specified Model

Convergence Visualization



Query Complexity for Detection

- Detecting $p = 0.1$ requires ~1,000 queries
- Detecting $p = 0.01$ requires ~100,000 queries
- Detecting $p = 0.001 \rightarrow$ query-prohibitive (millions+)

← Harder to detect

p (substitution rate)

Software-Only Auditing Methods Are Fundamentally Unreliable

Three failure modes systematically undermine every software-based detection approach

Method Type	Core Weakness	Why It Fails	Practical Limitation
Text-Based Statistical Tests	Query budget too large; subtle substitutions produce overlapping distributions	MMD and other text-output tests need thousands of queries and still produce false negatives against quantized or randomized substitutions	At p=0.01 substitution rate, statistical tests require >100K queries — economically prohibitive for auditors
Log-Probability Fingerprinting	Defeated by inference nondeterminism across software & hardware	transformers, vLLM, A100, and H100 all produce measurably different token log-probs for the same model, making substitution indistinguishable from infrastructure variation	No stable reference distribution exists in production environments — baseline drift makes any threshold unreliable
Adversarial Countermeasures	Providers can actively evacuate known benchmarks and randomize routing	Benchmark evasion (detecting prompt来源) and randomized substitution at low p make all passive detection fundamentally query-prohibitive	Auditors cannot detect when known benchmarks are selectively routed away from substitute models in real-time



No combination of software-only methods can provide provable detection guarantees under adversarial providers. All three failure modes are independent and combinable.

Trusted Execution Environments Provide Provable Cryptographic Model Integrity

Hardware-level attestation guarantees correct model weights and inference code without ZKP overhead

How TEE Attestation Works

Step 1 — Remote Attestation

The TEE generates a signed attestation quote binding the hardware state, loaded model weights (via hash), and inference code to a verifiable report.

Step 2 — Weight Integrity

Only the exact model weights cryptographically hashed in the attestation can produce the expected outputs. A substituted model produces a different hash.

Step 3 — Execution Integrity

The TEE enforces that inference runs exclusively within the enclave — no intermediate weights, no routing logic, no benchmark evasion is possible.

Step 4 — Client Verification

The client receives and verifies the attestation quote before accepting any inference result — providing cryptographic proof, not statistical confidence.

TEE Properties vs. Software Methods

Provable

Cryptographic proof of model identity, not statistical inference. Attestation quote is verifiable by any client.

Efficient

Modest performance overhead (~5–15%) vs. ZKPs which are computationally prohibitive for large models today.

Robust

Immune to inference nondeterminism, benchmark evasion, and randomized routing — hardware enforces execution integrity.

Deployable

NVIDIA confidential computing on H100/H200 GPUs provides production-grade TEE support today; no theoretical breakthroughs required.



NVIDIA Confidential Computing (H100/H200) provides the most practical TEE deployment path for LLM APIs today. The ecosystem is maturing rapidly — TEE-based model attestation is a near-term feasibility, not a research question.

Key Takeaways: Close the Verification Gap with Hardware Attestation

Evidence-driven conclusions for ML researchers, AI security practitioners, and cloud infrastructure engineers

① Model substitution is economically motivated and practically undetectable via software

FP8 quantization enables ~40% cost reduction with benchmark drops within error bars. Providers have strong financial incentives and technically easy paths to covert substitution.

② FP8 quantization barely affects benchmarks, masking substitution

Across five model families and four benchmarks, FP8 vs. FP16 differences are typically <2% — indistinguishable from normal variance without a reference model and large query budget.

③ Inference nondeterminism kills log-probability fingerprinting

Different transformers versions, vLLM versions, and GPU hardware (A100 vs. H100) produce measurably different token log-probs for the same model. No stable baseline exists in production.

④ Randomized substitution defeats all statistical tests

At substitution rate $p=0.01$, detecting a mixed distribution requires >100K queries. At $p=0.001$, detection is query-prohibitive. Providers can tune p to remain permanently below statistical detectability thresholds.

⑤ TEEs are the only currently viable path to verifiable LLM APIs

Hardware attestation provides cryptographic proof of model identity with modest performance overhead. Unlike ZKPs (impractical for large models today), TEE deployment on NVIDIA H100/H200 is available now.

Call to Action → Demand TEE attestation from LLM API providers. Adopt NVIDIA Confidential Computing. Build audit tooling on cryptographic integrity, not statistical inference.